# Jessica Kristen Rapson

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## EDUCATION

**Department of Statistics, University of Oxford** **Oxford, UK**
*Master of Science in Statistical Science (Merit)* *October 2023*

Thesis: *Improving Spatio-Temporal Forecasting by Extracting Indicators Based on Past Observations*

- Supervised by Sir Bernard Silverman, former Chief Scientific Adviser to the Home Office

**Munk School of Global Affairs and Public Policy, University of Toronto** **Toronto, CA**
*Master of Public Policy (3.98/4.00 CGPA)* *September 2021*

Thesis: *Forecasting the Post-Pandemic Public Transit Recovery in Toronto*

- Thesis used by INFC to support permanent public transit funding, presented to Deputy Minister of Infrastructure
- 2021 Valedictorian, gave speech at graduation ceremony and featured on Munk School website

**University of Toronto** **Toronto, CA**
*Honours Bachelor of Science, High Distinction (3.86/4.00 CGPA)* *June 2019*

- Dean's List Scholar, July 2016, July 2017, July 2018, and July 2019

**Relevant Coursework** Statistical Machine Learning (SB2.2), Advanced Topics in Statistical Machine Learning (SC4), Advanced Simulation Methods (SC5), Applied and Computational Statistics (SB1), Statistical Programming (SB2.0), Algorithmic Foundations of Learning (SC3), Panel Data Methods (PPG2010), Forecasting Models and Econometric Methods (RSM2129), Program Evaluation (PPG1008), Network Analysis (SC2)

## PUBLICATIONS

Rapson, J., Stephens, L., Maidment, R., et al. (2025) [DRAFT]. INFLOW-AI for predicting the onset and decay of unprecedented seasonal flooding using hybrid models: Applications for anticipatory action in South Sudan.

Rapson, J. McLaughlin, P.A., Wong, M.W., Morris, J., et al. (2025). Rules as Code for a More Transparent and Efficient Global Economy. Centre for International Governance Innovation.

Rapson, J. Radsch, C., Kumar, B., & Barop, J. (2025). A G7 Strategy for AI Competition and Consumer Rights. Centre for International Governance Innovation.

Rapson, J. & Kirton, J. (2020). Raising Compliance with G20 Commitments. *Global Solutions*, *5*(1).

Rapson, J. (2020). Increasing the Impact of the G7. *The Global Governance Project*.

Rapson, J., Kumar, B., Radsch, C., et al. (2025). Rules as Code for a More Transparent and Efficient Global Economy. *Think 7 Canada Task Force 2: Digitalization of the Global Economy*.

Rapson, J.,McLaughlin, P.A., Wong, MW., et al. (2025). A G7 Strategy for AI Competition and Consumer Rights. *Think 7 Canada Task Force 1: Transformative Technologies — AI and Quantum*.

Kirton, J., Warren, B., & Rapson, J. (2021). Creating Compliance with G20 and G7 Climate Change Commitments through Global, Regional and Local Actors .

Rapson, J. & Asrani, A. (2020). Ontario's tax system hurts marginal earnings of low-income individuals. *Public Policy and Governance Review*.

Palider, K., Patton, P., Barseghyan, H., [et al., including Rapson, J.]. (2021). A Diagrammatic Notation for Visualizing Epistemic Entities and Relations. *Scientonomy, Vol. 4*.

Rapson, J. (2024). How to foretell a flood: Predicting floods using machine learning. *Medium*.

Gerlein-Safdi, C., Pu, T., Bonifacio, R., [et al., including Rapson, J.]. (2024). Capturing and predicting flooding in the Sudd wetland: an integrative approach. *AGU24*.

Anderson, B., Slater, L., Rapson, J., et al. (2024). Inter-annual and long-term variability in streamflow elasticity to precipitation reveal bias in estimates of hydrological sensitivity. *EGU General Assembly Conference Abstracts*.

## PROFESSIONAL EXPERIENCE

**Algorithmic Governance Foundation** — **Oxford, UK**
*Director (Executive & Technical Lead)* — *May 2024 - Present*
- Director of an Oxford-based student research group that implements algorithmic solutions to key public sector problems
- Wrote two policy briefings for the G7 and presented at the French Ministry of Foreign Affairs on algorithmic governance

**University of Reading (Department of Meteorology)** — **Reading, UK**
*Visiting Scientist* — *July 2024 - Present*
- Received £20,000 grant from Google to build deployable machine learning model to predict flooding in South Sudan
- Built neural network for INFLOW project and provided live flood forecasts during the 2024 flood season to aid agencies
- Presented to UNHCR, ECMWF, and RC3 Long Night of Research; travelling to Kenya in June to present to local hydrologists

**Red Cross Red Crescent Climate Centre** — **The Hague, NL**
*Consultant* — *June 2024 - June 2024*
- Developed StormSpyder, a webscraping tool for monitoring tropical storms, used to reduce disaster aid response times

**Oxford Future Impact Group** — **Oxford, UK**
*Lead Researcher* — *September 2023 - May 2024*
- Led seven research projects with 36 students to develop algorithmic solutions for issues in sustainable global development
- Developed machine learning model to improve crop yields in Bihar, produced open source maps of public transit in Kenya, deployed web-scraping LLMs to detect anti-microbial resistance risk in India, produced simulations of climate policy impacts, digitised tax policies for G20 countries, built neural network to predict water supply in the Western U.S.

**Government of the United Kingdom (Foreign, Commonwealth and Development Office)** — **London, UK**
*Consultant* — *March 2024 - April 2024*
- Wrote report on "Artificial Intelligence-Driven Flood Forecasting" to steer department policy on the use of AI

**G7/20 Research Group** — **Toronto, CA**
*Senior Researcher/Statistician* — *August 2019 - Present*
- Uses machine learning models to predict and understand G7/G20 compliance with commitments made at summits
- Built prediction tool used by Sherpas at the 2023 Delhi Summit; presented tool to key diplomats at G20 Leader's Summit

**Government of Canada (Infrastructure Canada)** — **Ottawa, CA**
*Analyst, Data and Analytics* — *May 2020 - May 2021; September 2021 - October 2022*
- Developed forecasting model of post-pandemic public transit ridership, resulting in permanent public transit funding
- Produced nation-wide map of geospatial public transit data for use in analysis and consumer navigation tools
- Provided input resulting in department-wide investments to collect data on lead contamination and flood hazards

**Government of Canada (Canada Border Services Agency)** — **Ottawa, CA**
*Economic Analyst, Data Science and Advanced Analytics* — *May 2021 - September 2021*
- Worked on a large data science team to improve predictive machine learning model for border crossings

**PEARL Research Group** — **Toronto, CA**
*Researcher* — *October 2019 - July 2020*
- Developed models to determine whether individuals can successfully predict their job automation likelihood

**BFO Toronto** — **Toronto, CA**
*Data Management Specialist* — *July 2019 - August 2019*
- Used Python (NumPy, Pandas, Matplotlib, Statsmodels) to conduct financial analysis on donor contributions
- Developed predictive models (Scikit-Learn) to predict donor behaviour and identify financial opportunities

**180 Degrees Consulting** — **Toronto, CA**

*Director* — *June 2018 - May 2019*

- Director at the Toronto branch of the largest student-run consultancy in the world, serving non-profit clients
- Managed four projects with a large governing body (60,000+ fee-paying members) and a smaller non-profit

**University of Toronto Jackman Scholars-in-Residence** — **Toronto, ON**
*Researcher* — *April 2019 - May 2019*

- Used UML (Unified Modelling Language) to model historic worldviews and systems of scientific beliefs
- Assisted in compiling a database of scientific theories, co-authored book chapter on visualizing worldviews

**Collaborative Replication and Education Project** — **Toronto, CA**
*Researcher* — *January 2019 - April 2019*

- Worked with team to replicate economic research, performed statistical analysis on novel data using SPSS

**BMO Wealth Management** — **Toronto, CA**
*Management Consulting Intern* — *May 2018 - August 2018*

- Worked for the client, the Bank of Montreal, to assist in testing wealth management automations and macros
- Wrote a briefing on the new Basel Committee margin requirements for non-centrally cleared derivatives

## SOCIETIES & ACTIVITIES

**AI for Good** — **Oxford, UK**
*AI for Good Young AI Leader* — *January 2025 - Present*

- Supporting the United Nation's global "AI for Good" platform by driving responsible AI innovation at the Oxford Hub

**Statistics Without Borders** — **Remote**
*Statistical Analyst/Client Acquisition Team* — *May 2021 - Present*

- Conducted geospatial analysis of food pantry database to improve data quality and identify program gaps for FoodFinder
- Built a predictive model of sex trafficking using geospatial Bayesian networks and causal graphs for Prevention Now

**United for Smart Sustainable Cities (U4SSC) Initiative** — **Remote**
*Working Group Expert Member* — *April 2025 - Present*

- Contributing to a global UN collaboration for empowering cities by developing frameworks for AI integration into cities

**Public Good Initiative** — **Toronto, CA**
*Consultant and Data Analyst* — *December 2019 - July 2020*

- Worked with CNIB to research impact of assistive technology on the education of visually-impaired students

## TECHNICAL PROFICIENCIES

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|---|---|
| **Modeling & Analysis** | Python, R (data wrangling, modeling) • QGIS, Remote Sensing (spatial analysis) |
| **Visualization & Reports** | Tableau, R Shiny (dashboards) • Python (matplotlib, seaborn) • LaTeX |
| **Programming & Deployment** | AWS (EC2, S3, Lambda) • Git • Web Scraping (BeautifulSoup, Selenium) |
| **Generative ML Models** | LLaMA/Alpaca • LoRA (fine-tuning) • HPC/SLURM (distributed training) |
| **Languages** | English (Native) • French (CEFR C1) • Mandarin (HSK 3) |

## SELECT AWARDS & ACHIEVEMENTS

- **Ontario Graduate Scholarship** ($10,000), June 2020
- **Jackman Scholars-in-Residence Scholarship** ($1,000 + Research Fellowship), May 2019
- **Munk School TAship Merit Award** (~$3,000 + TAship), July 2020
- **First Place, SHIFT Public Sector Case Competition** (KPMG Job Offer), October 2019
- **Third Place, Zindi Digital Green Crop Yield Estimate Challenge** (€1,900), January 2024
- **Fraser-Crawford Scholarship** ($200), November 2018
- **University of Toronto Entrance Scholarship** ($2,000), September 2015
- **Dean's List Scholar**, July 2016, July 2017, July 2018, and July 2019